

📌 Part 1: Introduction to Decision Trees

◆ What is a Decision Tree?

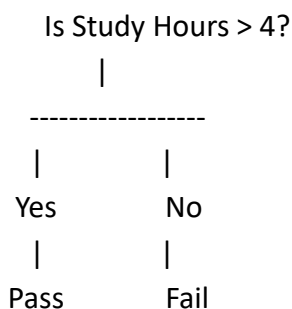
👉 A Decision Tree is a Supervised Machine Learning algorithm used for:

- Classification
- Regression

👉 It makes decisions by asking a series of questions.

◆ Why is it called a Decision Tree?

Because it looks like an upside-down tree:



The model reaches a decision by following branches.

📌 Part 2: Real-Life Example

Imagine a teacher wants to predict whether a student will pass.

Questions:

Study Hours > 4 ?

Yes → Pass

No → Fail

This question-based decision making is exactly how Decision Trees work.

Part 3: Theory Answers (IMPORTANT)

◆ 1. What is a Decision Tree?

👉 A Decision Tree is a supervised learning algorithm that makes predictions using decision rules.

◆ 2. Is Decision Tree Used for Classification or Regression?

👉 It can be used for both classification and regression.

◆ 3. What is the Root Node?

👉 The first question in the tree.

Example:

Is Study Hours > 4?

◆ 4. What is a Branch?

👉 A path connecting one decision to another.

◆ 5. What is a Leaf Node?

👉 The final prediction/output.

Example:

Pass

Fail

Part 4: Important Terms

Term	Meaning
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Root Node	Starting point
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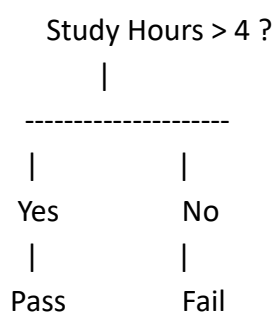
Term	Meaning
Branch	Decision path
Leaf Node	Final answer
Split	Dividing data
Tree Depth	Levels in tree

Part 5: Example Dataset

Study Hours Result

1	Fail
2	Fail
3	Fail
5	Pass
6	Pass
7	Pass

Part 6: Visual Tree Representation



Part 7: Install Required Library

`pip install scikit-learn`

Part 8: Import Libraries

```
import pandas as pd
```

```
from sklearn.tree import DecisionTreeClassifier
```

Part 9: Create Dataset

```
data = {  
    "Hours":[1,2,3,5,6,7],  
    "Result":[0,0,0,1,1,1]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Meaning

Value Meaning

0 Fail

1 Pass

Part 10: Features and Labels

```
X = df[["Hours"]]
```

```
y = df["Result"]
```

 X = Feature

 y = Label

Part 11: Create Decision Tree Model

```
model = DecisionTreeClassifier()
```

Part 12: Train the Model

```
model.fit(X, y)
```

👉 The model learns decision rules from data.

📌 Part 13: Make Prediction

```
prediction = model.predict([[4]])
```

```
print(prediction)
```

Possible Output:

```
[0]
```

Meaning:

👉 FAIL

📌 Part 14: Predict Multiple Values

```
prediction = model.predict([[2],[5],[8]])
```

```
print(prediction)
```

Possible Output:

```
[0 1 1]
```

📌 Part 15: Complete Program

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```

```
data = {
    "Hours": [1,2,3,5,6,7],
    "Result": [0,0,0,1,1,1]
}
```

```
df = pd.DataFrame(data)
```

```
X = df[["Hours"]]
```

```
y = df["Result"]
```

```
model = DecisionTreeClassifier()
```

```
model.fit(X, y)
```

```
prediction = model.predict([[5]])
```

```
print("Prediction:", prediction)
```

✦ Part 16: Advantages of Decision Tree

- ✓ Easy to understand
 - ✓ Easy visualization
 - ✓ Handles classification and regression
 - ✓ No complex mathematics
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✦ Part 17: Disadvantages of Decision Tree

- ✗ Can overfit data
 - ✗ Sensitive to small data changes
 - ✗ Large trees become complex
-

✦ Part 18: Real-World Applications

Education

- 👉 Student Performance Prediction
-

Banking

- 👉 Loan Approval Prediction
-

Healthcare

👉 Disease Prediction

E-Commerce

👉 Customer Classification

Security

👉 Fraud Detection

📌 Part 19: Decision Tree vs KNN

Decision Tree

KNN

Learns decision rules Uses nearest neighbors

Fast prediction Slower prediction

Easy visualization No tree structure

Good interpretability Less interpretable

📌 Part 20: Overfitting in Decision Trees

◆ What is Overfitting?

👉 When a model memorizes training data instead of learning patterns.

Result:

- Excellent training performance
 - Poor performance on new data
-

Solution

Limit tree depth:

```
model = DecisionTreeClassifier(max_depth=3)
```

Part 21: Decision Tree Workflow

Dataset



Features (X)



Decision Rules



Tree Creation



Prediction

Tasks for Day 40

Task 1: Theory

1. What is a Decision Tree?
 2. What is a Root Node?
 3. What is a Branch?
 4. What is a Leaf Node?
 5. Advantages of Decision Trees?
-

Task 2: Dataset Creation

1. Create student result dataset
 2. Create loan approval dataset
 3. Create disease prediction dataset
-

Task 3: Model Building

1. Create Decision Tree model
 2. Train model
 3. Make predictions
-

Task 4: Prediction Practice

Predict:

1. Student result
 2. Disease status
 3. Loan approval
-



Task 5: Tree Analysis

1. Identify root node
 2. Identify leaf nodes
 3. Explain decision path
-



Task 6: Compare Algorithms

Compare:

1. Logistic Regression
2. KNN
3. Decision Tree